

Solution set 2

Problem 1:

$$e^{i\theta(\hat{n}\cdot\vec{\sigma})/2} \vec{\sigma} e^{-i\theta(\hat{n}\cdot\vec{\sigma})/2}$$

$$= (\cos\theta/2 + i(\hat{n}\cdot\vec{\sigma})\sin\theta/2) \vec{\sigma} (\cos\theta/2 - i(\hat{n}\cdot\vec{\sigma})\sin\theta/2)$$

$$= \vec{\sigma} \cos^2\theta/2 + \sin^2\theta/2 (\hat{n}\cdot\vec{\sigma}) \vec{\sigma} (\hat{n}\cdot\vec{\sigma}) + i[(\hat{n}\cdot\vec{\sigma}) \vec{\sigma} - \vec{\sigma} (\hat{n}\cdot\vec{\sigma})] \sin\theta/2 \cos\theta/2 \quad \text{--- (1)}$$

now.

$$\begin{aligned} (\hat{n}\cdot\vec{\sigma}) \sigma_k - \sigma_k (\hat{n}\cdot\vec{\sigma}) &= \left(\sum_j n_j \sigma_j\right) \sigma_k - \sigma_k \left(\sum_j n_j \sigma_j\right) \\ &= \sum_j n_j (\sigma_j \sigma_k - \sigma_k \sigma_j) \\ &= 2i \sum_j n_j \epsilon_{jkl} \sigma_l \\ &= -2i(\hat{n} \times \vec{\sigma})_k \end{aligned}$$

$$[(\hat{n}\cdot\vec{\sigma}) \cdot \vec{\sigma} - \vec{\sigma} (\hat{n}\cdot\vec{\sigma})] = -2i(\hat{n} \times \vec{\sigma}) \quad \text{--- (2)}$$

Also

$$\begin{aligned} (\hat{n}\cdot\vec{\sigma}) \sigma_k (\hat{n}\cdot\vec{\sigma}) &= \sum_{j,l} n_j \sigma_j \sigma_k n_l \sigma_l \\ &= \sum_{j,l} n_j n_l \sigma_j \sigma_k \sigma_l \\ &= \sum_{j,l} n_j n_l \{ \delta_{jk} + i \sum_m \epsilon_{jkm} \sigma_m \} \sigma_l \\ &= n_k (\hat{n}\cdot\vec{\sigma}) + i \sum_{j,l,m} n_j n_l \epsilon_{jkm} \sigma_m \sigma_l \end{aligned}$$

$$= \eta_k(\hat{n} \cdot \vec{\sigma}) + i \sum_{jlm} n_j n_l \epsilon_{jkm} \left\{ \delta_{ml} + i \sum_n \epsilon_{men} \sigma_n \right\}.$$

$$= \eta_k(\hat{n} \cdot \vec{\sigma}) + i \sum_{jle} \underbrace{n_j n_l \epsilon_{jke}}_{\substack{\text{Symmetric} \\ \text{in } j, l}} - \sum_{jlmn} n_j n_l \epsilon_{jkm} \epsilon_{men} \sigma_n$$

$\downarrow$ Symmetric in j, l $\downarrow$ Antisymmetric in j, l
 $\searrow$ zero!

$\epsilon_{jkm} = \epsilon_{mjk}.$

$$= \eta_k(\hat{n} \cdot \vec{\sigma}) - \sum_{jlmn} n_j n_l \underbrace{\epsilon_{mjk} \epsilon_{men}}_{(\delta_{je} \delta_{kn} - \delta_{jn} \delta_{ke})} \sigma_n.$$

$$= \eta_k(\hat{n} \cdot \vec{\sigma}) - \sigma_k + \eta_k(\hat{n} \cdot \vec{\sigma}).$$

$$= 2\eta_k(\hat{n} \cdot \vec{\sigma}) - \sigma_k.$$

$$\therefore (\hat{n} \cdot \vec{\sigma}) \vec{\sigma} (\hat{n} \cdot \vec{\sigma}) = 2\vec{n}(\hat{n} \cdot \vec{\sigma}) - \vec{\sigma}. \quad \text{--- (3)}$$

Substituting (2) and (3) in eq. (1) we get-

$$e^{i\theta(\hat{n} \cdot \vec{\sigma})/2} \vec{\sigma} e^{-i\theta(\hat{n} \cdot \vec{\sigma})/2}$$

$$= \vec{\sigma}(\cos^2 \theta/2 - \sin^2 \theta/2) + 2(\hat{n} \cdot \vec{\sigma}) \hat{n} \sin^2 \theta/2$$

$$+ 2(\hat{n} \times \vec{\sigma}) \sin \theta/2 \cos \theta/2$$

$$= [\vec{\sigma} - \vec{n}(\hat{n} \cdot \vec{\sigma})] \cos \theta + (\hat{n} \times \vec{\sigma}) \sin \theta + (\hat{n} \cdot \vec{\sigma}) \hat{n}$$

Where we used the fact that-

$$\cos \theta = \cos^2 \theta/2 - \sin^2 \theta/2$$

$$1 = \sin^2 \theta/2 + \cos^2 \theta/2, \quad \sin \theta = 2 \sin \theta/2 \cos \theta/2$$

Problem 2 The Hamiltonian can be written as

$$\begin{aligned}
 H &= \begin{pmatrix} E_1 & \gamma e^{i\omega t} \\ \gamma e^{-i\omega t} & E_2 \end{pmatrix} \\
 &= \begin{pmatrix} \frac{E_1+E_2}{2} & 0 \\ 0 & \frac{E_1+E_2}{2} \end{pmatrix} + \begin{pmatrix} \frac{E_1-E_2}{2} & \gamma e^{i\omega t} \\ \gamma e^{-i\omega t} & -\frac{E_1-E_2}{2} \end{pmatrix} \\
 &= \frac{E_1+E_2}{2} \mathbb{1} + \frac{E_1-E_2}{2} \sigma_3 + \gamma [\cos \omega t \sigma_1 - \sin \omega t \sigma_2] \\
 &= e^{\frac{i\omega t}{2} \sigma_3} \left[\frac{E_1+E_2}{2} \mathbb{1} + \frac{E_1-E_2}{2} \sigma_3 + \gamma \sigma_1 \right] e^{-\frac{i\omega t}{2} \sigma_3}
 \end{aligned}$$

We know that-

$$i\hbar \frac{\partial}{\partial t} \begin{pmatrix} C_1(t) \\ C_2(t) \end{pmatrix} = H \begin{pmatrix} C_1(t) \\ C_2(t) \end{pmatrix}.$$

$$\mathcal{E}_0 = \frac{E_1+E_2}{2}$$

$$\mathcal{E}_3 = \frac{E_1-E_2}{2}$$

Let us define $\begin{pmatrix} \tilde{C}_1(t) \\ \tilde{C}_2(t) \end{pmatrix} = e^{\frac{-i\omega t}{2} \sigma_3} \begin{pmatrix} C_1(t) \\ C_2(t) \end{pmatrix} \quad \text{--- (1)}$

Then we see that-

$$i\hbar \frac{\partial}{\partial t} \begin{pmatrix} \tilde{C}_1(t) \\ \tilde{C}_2(t) \end{pmatrix} = \frac{\hbar\omega}{2} \sigma_3 \begin{pmatrix} \tilde{C}_1(t) \\ \tilde{C}_2(t) \end{pmatrix} + (\mathcal{E}_0 + \mathcal{E}_3 \sigma_3 + \gamma \sigma_1) \begin{pmatrix} \tilde{C}_1(t) \\ \tilde{C}_2(t) \end{pmatrix}$$

$$\Rightarrow i\hbar \frac{\partial}{\partial t} \begin{pmatrix} \tilde{C}_1(t) \\ \tilde{C}_2(t) \end{pmatrix} = \left[\mathcal{E}_0 + \left(\mathcal{E}_3 + \frac{\hbar\omega}{2} \right) \sigma_3 + \gamma \sigma_1 \right] \begin{pmatrix} \tilde{C}_1(t) \\ \tilde{C}_2(t) \end{pmatrix}.$$

Hence we know

$$\begin{pmatrix} \tilde{C}_1(t) \\ \tilde{C}_2(t) \end{pmatrix} = \exp \left(\frac{-i}{\hbar} \left[\mathcal{E}_0 + \left(\mathcal{E}_3 + \frac{\hbar\omega}{2} \right) \sigma_3 + Y\sigma_1 \right] t \right) \begin{pmatrix} \tilde{C}_1(0) \\ \tilde{C}_2(0) \end{pmatrix}$$

We are given

$$C_1(0) = 1 \quad C_2(0) = 0.$$

$$\therefore \tilde{C}_1(0) = 1 \quad \tilde{C}_2(0) = 0 \quad \text{using eq. (1).}$$

$$\begin{pmatrix} \tilde{C}_1(t) \\ \tilde{C}_2(t) \end{pmatrix} = e^{\frac{-i}{\hbar} \mathcal{E}_0 t} \begin{pmatrix} \cos \Omega t & -i \hat{n} \cdot \vec{\sigma} \sin \Omega t \end{pmatrix} \begin{pmatrix} \tilde{C}_1(0) \\ \tilde{C}_2(0) \end{pmatrix}$$

$$\text{Here } \Omega = \sqrt{\left(\frac{\mathcal{E}_3}{\hbar} + \frac{\omega}{2} \right)^2 + \frac{Y^2}{\hbar^2}} \quad \text{--- (2)}$$

$$n_1 = \frac{Y}{\hbar \Omega} \quad \text{--- (3)}$$

$$\text{and } \hat{n} = \frac{1}{\Omega} \left\{ \frac{Y}{\hbar} \hat{i} + \left(\frac{\mathcal{E}_3}{\hbar} + \frac{\omega}{2} \right) \hat{k} \right\} \quad n_3 = \left(\frac{\mathcal{E}_3}{\hbar} + \frac{\omega}{2} \right) \frac{1}{\Omega} \quad \text{--- (4)}$$

$$\begin{pmatrix} \tilde{C}_1(t) \\ \tilde{C}_2(t) \end{pmatrix} = e^{\frac{-i}{\hbar} \mathcal{E}_0 t} \begin{pmatrix} \cos \Omega t - i n_3 \sin \Omega t \\ -i n_1 \sin \Omega t \end{pmatrix}$$

$$\therefore \begin{pmatrix} C_1(t) \\ C_2(t) \end{pmatrix} = e^{\frac{-i}{\hbar} \mathcal{E}_0 t} \begin{pmatrix} e^{\frac{i\omega t}{2}} (\cos \Omega t - i n_3 \sin \Omega t) \\ -i e^{\frac{-i\omega t}{2}} n_1 \sin \Omega t \end{pmatrix}$$

Probability to find the state in $|2\rangle$ is $n_1^2 \sin^2 \Omega t$.

n_1 and Ω were defined in eq. (2) and (3).

Problem 3

3-1

$$\vec{A} = \frac{B}{2} (-y \vec{i} + x \vec{j})$$

$$(1) \quad \vec{B} = \vec{\nabla} \times \vec{A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y \frac{B}{2} & x \frac{B}{2} & 0 \end{vmatrix} = 0 \vec{i} + 0 \vec{j} + \left(\frac{B}{2} + \frac{B}{2}\right) \vec{k} = B \vec{k}$$

$$(2) \quad \vec{F} = q \frac{\vec{v} \times \vec{B}}{c} \quad \text{Classical particle.}$$

$$F = q \frac{\omega_0 R B}{c} = \mu \omega_0^2 R$$

$$\Rightarrow \quad \omega_0 = \frac{qB}{\mu c}$$

$$(3) \quad H = \frac{(P_x + qyB/2c)^2}{2\mu} + \frac{(P_y - qxB/2c)^2}{2\mu}$$

$$\text{if let } Q = (cP_x + qyB/2)/qB$$

$$P = (P_y - qxB/2c)$$

$$[Q, P] = \left[\frac{c}{qB} P_x, -\frac{qB}{2c} x \right] + \left[\frac{1}{2} y, P_y \right]$$

$$= \frac{1}{2} i\hbar + \frac{1}{2} i\hbar = i\hbar$$

So Q and P are canonical.

$$\Rightarrow H = \frac{P^2}{2\mu} + \frac{g^2 B^2}{2\mu c^2} Q^2$$

$$\hookrightarrow \text{SHO with } \omega_0 = \frac{gB}{\mu c}$$

$$E_n = (n + \frac{1}{2}) \hbar \omega_0$$

Wave fun depend on Q

$$Q = (CP_x + \frac{g}{2} Y B / c) / \frac{g}{2} B$$

$-\infty < Y < \infty$ so degeneracy ∞

$$(4) \quad H = \frac{(P_x + \frac{g}{2} Y B / c)^2}{2\mu} + \frac{(P_y - \frac{g}{2} X B / c)^2}{2\mu}$$

$$= \frac{P_x^2 + P_y^2}{2\mu} + \frac{gB}{2\mu c} (Y P_x - X P_y) + \frac{g^2 B^2}{8\mu c^2} (x^2 + y^2)$$

$$= \underbrace{\frac{1}{2\mu} (P_x^2 + P_y^2) + \frac{g^2 B^2}{8\mu c^2} (x^2 + y^2)}_{2D \text{ SHO}} + \frac{gB}{2\mu c} L_z$$

$$\omega_0' = \frac{gB}{2\mu c} = \frac{\omega_0}{2}$$

$$= H(\frac{\omega_0}{2}, \mu) - \frac{\omega_0}{2} L_z$$

$$\therefore [H, L_z] = 0 \quad [H, L^2] = 0$$

$$[H(\frac{\omega_0}{2}, \mu) - \frac{\omega_0}{2} L_z, L_z] = 0$$

$$[H(\frac{\omega_0}{2}, \mu) - \frac{\omega_0}{2} L_z, L^2] = 0$$

$\Rightarrow$ the same basis that diagonalized $H(\frac{\omega_0}{2}, \mu)$ will diagonalized H .

from 12.3.7 (7)

$$E_{2DSD} = (2k + |m| + 1) \hbar \frac{\omega_0}{2} \quad (k=0, 1, 2, \dots)$$

m integer

$$E = (2k + |m| + 1) \hbar \frac{\omega_0}{2} - \frac{\omega_0}{2} m \hbar$$

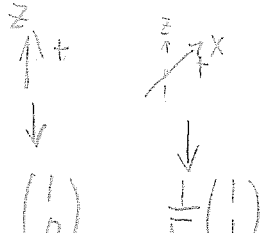
$$= \left(k + \frac{|m|}{2} - \frac{m}{2} + \frac{1}{2} \right) \hbar \omega_0$$

$$\frac{|m|}{2} - \frac{m}{2} = \begin{cases} 0 & m = \text{positive integer} \\ |m| & m = \text{negative integer} \end{cases}$$

So $E = \left(n + \frac{1}{2} \right) \hbar \omega_0$ $n = 0, 1, 2, \dots$

Problem 4

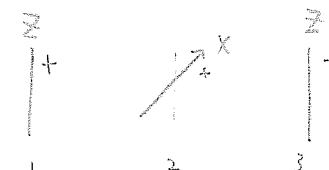
(1)



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$P = \left| \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right|^2 = \boxed{\frac{1}{2}}$$

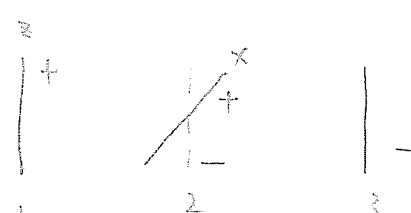
(2)



$$P_{12} = \frac{1}{2} \quad P_{23} = \left| \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right|^2 = \frac{1}{2}$$

$$P = P_{12} \cdot P_{23} = \boxed{\frac{1}{4}}$$

(3)



$$P_{12}^+ = \frac{1}{2} \quad P_{12}^- = \frac{1}{2}$$

$$P_{23}^+ = \left| \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right|^2 = \frac{1}{2} \quad P_{23}^- = \frac{1}{2}$$

$$P = P_{12}^+ P_{23}^+ + P_{12}^- P_{23}^- = \boxed{\frac{1}{2}}$$